# SRIJAN SRIVASTAVA

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## EDUCATION

**Vellore Institute of Technology** — **Vellore, India**
B.Tech in Computer Science and Engineering, CGPA 8.35/10.0 – Expected July 2027 — July 2023 – Present

**Little Flower School** — **India**
ISC – 89% — July 2021 – July 2022

**Little Flower School** — **India**
ICSE – 86% — July 2019 – July 2020

## TECHNICAL SKILLS

**Languages**: Java, Python, C++, C, Go, TypeScript, JavaScript, SQL, Bash

**Core CS**: Data Structures & Algorithms, Object-Oriented Design, System Design, Distributed Systems, Complexity Analysis

**Reliability & Observability (SRE)**: Prometheus, Grafana, PagerDuty, SLIs/SLOs/SLAs, Error Budgets, Incident Command, Blameless Post-Mortems, Chaos Engineering, Linux Internals (systemd, cgroups, strace), Networking (TCP/IP, DNS, Load Balancing), Terraform, Capacity Planning, On-Call Rotation

**AI & ML**: PyTorch, TorchScript, PyTorch Distributed, Custom Training Loops, LLM Fine-Tuning, LoRA/QLoRA, PEFT, SFT, Tokenization, BPE, SentencePiece, Hugging Face Transformers, Neural Network Optimization, Hyperparameter Tuning, Transfer Learning, Quantization, Computer Vision (YOLOv8), RAG, Vector Search, Embeddings, LangChain, LangGraph, Agentic Workflows, MLOps, Model Inference & Deployment

**Testing & Quality Engineering**: Selenium, Cypress, Playwright, JUnit, TestNG, PyTest, REST Assured, Appium, JMeter, Jenkins, Test Architecture, API Testing, UI Automation, Load Testing, Performance Profiling, TDD/BDD, Regression Suites, Regression Testing, CI/CD Quality Gates

**Data & Business Analysis**: SQL, ETL, Tableau, Power BI, Pandas, Excel, KPI Reporting, Dashboard Creation, Exploratory Data Analysis, Cohort & Trend Analysis, A/B Testing, Data Storytelling, Business Impact Analysis, Requirements Gathering, BRD, Functional Specifications, Process Documentation, Use Cases, RACI Matrix, Requirements Traceability Matrix, UAT, Stakeholder Interviews, Gap Analysis, Process Mapping (BPMN), Business Process Improvement, User-Story Creation, JIRA, Confluence, Visio

**Backend & Cloud**: Spring Boot, Node.js, Next.js, React, Express.js, FastAPI, REST APIs, WebSockets, Microservices, PostgreSQL, MySQL, MongoDB, CockroachDB, Redis, AWS, Azure, GCP, Docker, Kubernetes, GitHub Actions, Linux, Agile/Scrum, Git, GitHub

## EXPERIENCE

**ExpLocal** — **Vellore, India**
Frontend Engineer Intern — Aug 2025 – Oct 2025

- Developed responsive and scalable frontend interfaces for a **travel marketplace platform**, building reusable UI components using component-based architecture and improving page-load performance by approximately **25%**.
- Integrated **REST APIs** and optimized frontend state management to support seamless travel discovery and booking workflows while maintaining responsive behavior across devices.
- Collaborated with backend and design teams to implement consistent **UI/UX**, translate requirements into reusable components, and reduce code redundancy across the application.
- **Tech Stack**: React.js, Tailwind CSS, JavaScript, TypeScript, Git, MySQL

**Lok Mitra Venture** — **Remote**
Business Analyst – Business Analysis Internship — 2026

- Conducted **requirements analysis** from business briefs, translating stakeholder needs into **process documentation, functional specifications, and structured user requirements** while maintaining traceability between requirements and acceptance criteria.
- Analysed business datasets using **SQL and Excel**, developed **KPI-focused dashboards**, and communicated findings to non-technical stakeholders through concise recommendations focused on business impact and actionable improvements.

## PROJECTS

**FITOS – AI-Powered Personal Fitness Operating System** – *Flutter, TypeScript, Fastify, PostgreSQL, Gemini, Docker, Cloud Run* **2026**

- Architected a **full-stack AI fitness platform** using Flutter, TypeScript/Fastify, PostgreSQL and Firebase, implementing adaptive workout programming, progressive overload, nutrition tracking, recovery insights, health data integration, and progress analytics.
- Built a controlled **Gemini tool-calling architecture** with authenticated user context and deterministic backend services for workout calculations, progression, exercise substitution, deload detection, and personalized recommendations; implemented **offline-first synchronization** with local persistence, retries, idempotency, and conflict recovery.
- Integrated **Android Health Connect** and deployed containerized backend services using Docker, Google Cloud Run, Artifact Registry, Secret Manager, Neon PostgreSQL, and GitHub Actions with automated migrations, health checks, CI/CD validation, and rollback support.

**L2 Escape Bench – Forced-Inclusion Experiment Harness** – *Python, Ethereum, Arbitrum, OP Stack, Sepolia* **2026**

- Developed a reproducible **Layer-2 forced-inclusion and escape experiment harness** for evaluating transaction recovery mechanisms across **OP Sepolia and Arbitrum Sepolia**.
- Implemented CLI-driven experiment workflows for **transaction submission, RPC interaction, environment-based configuration, experiment validation, and automated test execution**, enabling repeatable L2 escape-hatch analysis.
- Investigated the relationship between **L1 transaction submission and L2 inclusion mechanisms**, designing controlled experiments to evaluate transaction execution when normal L2 sequencing is unavailable.

**Pulse – Desktop Performance Optimization Platform** – *Electron, React, C#, ASP.NET Core* **2026**

- Developed a desktop performance optimization platform using a **Reason–Act architecture** that reacts to real-time system load and dynamically activates performance profiles when resource-intensive applications and games are launched.
- Built an **Electron + React** desktop interface with an **ASP.NET Core Minimal API** backend in C#, enabling process-priority management, Windows power-profile switching, and automated control of user-selected background processes.

**Adaptive Multi-Biosignal Emotion Detection System** – *Embedded C, ESP32, Signal Processing, IoT* **2026**

- Designed a real-time **multi-biosignal emotion detection system** for resource-constrained embedded hardware, processing heart-rate, motion, and skin-conductance signals using a deterministic architecture with less than **600 KB SRAM**.
- Developed an **Adaptive Weighted Fusion (AWF)** architecture that dynamically adjusts sensor trust based on signal quality, providing graceful degradation when individual biosensor channels become unreliable.
- Implemented **Emotional Trajectory Prediction (ETP)**, **Stress Accumulation Integration (SAI)**, and **Session Wellness Reporting (SMHR)** modules for proactive stress detection, temporal stress-load modelling, haptic intervention, and embedded wellness assessment.
- Integrated real-time physical feedback through **haptic actuation, visual indicators, display, and wireless monitoring dashboard**; developed the invention as a **patent application with VIT Vellore**.

## LEADERSHIP & EXTRACURRICULAR ACTIVITIES

**Association for Computing Machinery, VIT** **Vellore, India**
Senior Inner Core Member 2024 – Present

- Organized **Cryptic Hunt '24**, VIT's flagship app-based treasure hunt event, contributing to event planning, technical coordination, and execution.

**IEEE Computer Society, VIT** **Vellore, India**
Core Member 2024 – Present

- Organized **Hack Battle '24**, a flagship hackathon with **300+ participants**, and mentored **20 peers** in web development to support technical growth and project execution.

## CERTIFICATIONS

**IBM DevOps and Software Engineering Professional Certificate**